

Autocovariance for the ARMA(1, 1) Process

We have

$$\gamma(0) = \phi_1 \gamma(1) + \sigma_z^2 (1 + \theta_1 (\phi_1 + \theta_1)) \quad (1)$$

$$\gamma(1) = \phi_1 \gamma(0) + \theta_1 \sigma_z^2 \quad (2)$$

Substitute (2) into (1)

$$\begin{aligned} \gamma(0) &= \phi_1 [\phi_1 \gamma(0) + \theta_1 \sigma_z^2] \\ &\quad + \sigma_z^2 (1 + \theta_1 (\phi_1 + \theta_1)) \\ &= \phi_1^2 \gamma(0) + \sigma_z^2 [\phi_1 \theta_1 + 1 + \theta_1 (\phi_1 + \theta_1)] \end{aligned}$$

$$\begin{aligned} \gamma(0) (1 - \phi_1^2) &= \sigma_z^2 [1 + 2\phi_1 \theta_1 + \theta_1^2] \\ \Rightarrow \gamma(0) &= \frac{\sigma_z^2}{1 - \phi_1^2} (1 + 2\phi_1 \theta_1 + \theta_1^2) \end{aligned}$$

Substitute the above into (2)

$$\begin{aligned} \therefore \gamma(1) &= \phi_1 \left[\frac{\sigma_z^2 (1 + 2\phi_1 \theta_1 + \theta_1^2)}{1 - \phi_1^2} \right] + \theta_1 \sigma_z^2 \\ &= \frac{\sigma_z^2}{1 - \phi_1^2} [\phi_1 (1 + 2\phi_1 \theta_1 + \theta_1^2) + \theta_1 (1 - \phi_1^2)] \\ &= \frac{\sigma_z^2}{1 - \phi_1^2} [\phi_1 + \phi_1^2 \theta_1 + \phi_1 \theta_1^2 + \theta_1] \end{aligned}$$

$$= \frac{\sigma_z^2}{1 - \phi_1^2} (\phi_1 + \theta_1)(1 + \phi_1 \theta_1)$$

What about $\gamma(h)$ for $|h| > 1$?

Consider $h > 1$

$$X_t - \phi_1 X_{t-1} = Z_t + \theta_1 Z_{t-1} \quad (*)$$

Multiply both sides of (*) by X_{t-h} and take the expectation of both sides.

$$\begin{aligned} \Rightarrow E(X_t X_{t-h}) - \phi_1 E(X_{t-1} X_{t-h}) \\ = E(Z_t X_{t-h}) + \theta_1 E(Z_{t-1} X_{t-h}) \end{aligned}$$

$$\Rightarrow \gamma(-h) - \phi_1 \gamma(-h+1) = 0 \quad (\text{causality})$$

$$\Rightarrow \gamma(h) - \phi_1 \gamma(h-1) = 0 \quad (\text{stationarity})$$

$$\Rightarrow \gamma(h) = \phi_1 \gamma(h-1)$$

$$\gamma(2) = \phi_1 \gamma(1)$$

$$\gamma(3) = \phi_1 \gamma(2) = \phi_1^2 \gamma(1)$$

$$\gamma(4) = \phi_1 \gamma(3) = \phi_1^3 \gamma(1)$$

Generally, for $|h| > 1$ $\gamma(h) = \phi_1^{|h|-1} \gamma(1)$.

$$\rho(h) = \text{Corr}(X_t, X_{t+h}) = \frac{\gamma(h)}{\gamma(0)}$$

$$\text{Corr}(X_t, X_{t+h}) = \frac{\text{Cov}(X_t, X_{t+h})}{\sqrt{\text{Var}(X_t) \text{Var}(X_{t+h})}}$$